

HBC RADIO

ATAK Plugin — Interface Control Document

Ham Binary CoT over Distance-Vector Mesh Radio

Version 1.5 | KE8TQB | github.com/ke8tqb/HBC-ATAK-Plugin

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1. Overview

HBC Radio is a plugin (an add-on) for ATAK, the Android Team Awareness Kit mapping app. It lets ATAK devices share map information — positions, markers, chat messages, and alerts — using ordinary handheld ham radios instead of the internet or a cell network. The phone plays specially coded sound into the radio; another phone hears that sound on its radio and turns it back into map data.

The system is built in three layers, like a stack of responsibilities. Each layer is specified in its own section of this document:

- **Physical layer (Section 3)** — the “voice” of the system: one of three selectable audio modems that turn bytes into sound and back.
- **Mesh network layer (Section 4)** — the “postal service”: it figures out who can hear whom and can pass a message through a middle station to reach someone out of range. Based on a published distance-vector routing design (see References).
- **Application layer (Section 5)** — the “language”: the HBC Protocol, which squeezes a map event into a few dozen bytes so it can be sent quickly over a slow radio link.

In plain terms: HBC (the language) writes a very short letter, the mesh (the postal service) addresses it and picks the route, and the modem (the voice) reads it out loud over the radio.

NOTE: Everything is transmitted unencrypted (in the clear), as required by amateur-radio rules (FCC Part 97). The operator’s callsign is inside every transmission, which satisfies station-identification requirements.

2. System Architecture

The path one message takes from your screen to the other station’s screen:

ATAK map event → HBC encoder (shrinks it to bytes) → MeshRouter (adds addressing) → modem (turns bytes into audio) → radio → ...air... → radio → modem → MeshRouter → HBC decoder → ATAK map

On receive, the mesh layer decides what to do with each frame: deliver it to this device, forward it toward its destination, rebroadcast it (announces), or ignore it.

Layer	Code that implements it	How things are addressed
Application	HbcEncoder / HbcDecoder	Callsign inside the HBC header; predictable UIDs (section 5.3)

Network	MeshRouter	Origin / Transmitter / Destination / Next-hop callsigns
Physical	AudioModem, OfdmModem, MercuryModem	None — just a byte pipe (AFSK adds a fixed AX.25 wrapper for station ID)

3. Physical Layer (Audio Modems)

A modem converts bytes into sound and sound back into bytes. HBC Radio offers three, each suited to different radios and conditions. All three carry the exact same mesh frame, but they are not compatible with each other — **every station on a channel must select the same modem** (Figure 4).

Parameter	AFSK1200	OFDM COFDM TV	Mercury HF
Waveform	Bell 202 AFSK, 1200 baud	QPSK OFDM (rattlegram “short”), 1600 Hz BW	FreeDV DATAC4 raw-data OFDM (codec2)
Sample rate	48000 Hz	48000 Hz	8000 Hz
Payload capacity per burst	Practical AX.25 limits (>200 B)	169 B	52 B (54 B frame minus framing)
Error protection	HDLC CRC-16 (detect only)	Polar code + CRC-32	LDPC + CRC-16
Best for	VHF/UHF FM; compatible with classic packet radio	VHF/UHF FM; survives phone audio processing	HF SSB; works at very weak signal levels

3.1 AFSK1200 / AX.25

The classic 1980s packet-radio sound (two alternating tones, 1200 bits per second). Mesh frames ride inside a single AX.25 “UI” frame. The AX.25 addresses are only wrapping paper — they are not used for routing: destination is always HBC, source is the operator’s ham callsign (which doubles as the legal station ID), and no digipeater path is used. Each transmission is rendered in advance as: lead-in silence (150 ms) → optional VOX leader tone → HDLC flag preamble (“TX Dwell,” default 500 ms) → the frame → tail silence (250 ms).

3.2 OFDM COFDM TV (rattlegram)

A modern, robust waveform that packs data onto many small carriers at once. Framing inside the 170-byte block: byte 0 = mesh frame length (1–169), bytes 1..n = the mesh frame, zero-padded. The operator callsign also rides in the waveform’s own metadata header. Audio carrier is 1500 Hz; the VOX leader is made of noise symbols (~180 ms each).

3.3 Mercury HF (FreeDV DATAC4)

Designed for long-range HF radio, where signals are weak and distorted. The burst layout matches the Mercury HF modem exactly: preamble → one DATAC4 data frame with a CRC-16 checksum in its last two bytes → postamble, with 100 ms of silence on each end. Framing inside the 54 usable bytes: byte 0 = length, byte 1 = the letter 'H' (a marker that says “this is HBC traffic”), bytes 2..n = the mesh frame, zero-padded.

NOTE: This speaks Mercury’s *waveform*, not Mercury’s higher-level ARQ data-link protocol. A desktop Mercury installation can demodulate these bursts but will not pass them to its TCP data port.

3.4 Channel Access — CSMA (legacy contention)

Only one station can talk on a radio channel at a time; two at once garble both (a “collision”). In CSMA mode (the default and the only mode before plugin 0.22), every modem *listens first*: incoming audio energy is compared against an automatically learned background-noise level, and the channel counts as busy when audio is 4× louder than that background (held for 400 ms past the last loud moment so brief gaps inside someone’s transmission don’t look like free air). When the channel goes quiet, the station still waits a random 150–550 ms — so two stations that were both waiting don’t key up at the same instant. After 8 seconds of continuous busy (15 s on Mercury) the frame is sent anyway so traffic can never be blocked forever. AFSK also uses its demodulator’s data-carrier-detect as a second “someone is talking” signal.

When to use which: CSMA is fine for 2–5 stations. It cannot see *hidden terminals* (two stations that cannot hear each other but both reach a third), and with VOX radios every frame pays the radio’s full key-up/hang cycle — so with many stations, collisions and dead air grow. For nets up to ~20 stations, select the **Ring MAC** (section 3.5) on every station instead.

3.5 Channel Access — Ring MAC (deterministic rotation)

New in plugin 0.22. Instead of competing for the channel, stations take **turns in a fixed, known order**: every station sorts the callsigns it knows about (its own plus the mesh address book of section 4.7, which fills itself just by listening) and rotates through that list. The list IS the schedule — there is no coordinator station, no “token” packet, and **no change to anything transmitted on the air**: the MAC only decides *when* a station may key up, never what the bytes look like. 0.21 and 0.22 stations therefore interoperate; a CSMA station in a Ring net is simply heard as carrier and deferred to.

A station transmits only during its own turn — and then sends **everything it has queued as one continuous keying** (one batch: single lead-in, frames back-to-back, single tail), so the radio’s VOX hang is paid once per turn instead of once per frame. Every station advances the turn pointer on its own, using whichever of these happens first:

Rule	Trigger	Purpose
1. Early release	The owner's carrier was heard and the channel has then stayed clear for Guard ms	Short messages finish early; Guard absorbs the radio's VOX hang so the tail is never mistaken for free air
2. Silent skip	No carrier within Skip ms of the turn starting	Idle stations cost ~1 s, not a whole turn
3. Deadline	Max turn + Guard ms since the turn started	Keeps the ring alive past hidden terminals and dead stations — even a station that hears nothing advances

Two additional behaviors keep every station's pointer agreeing: decoding any mesh frame **re-aligns the pointer to that frame's Transmitter callsign** (hard evidence beats timers), and a station never keys up while the channel is busy (interlock), even on its own turn. A joining station listens for one full cycle (at most 30 s) before taking its first turn; roster changes apply only when the rotation wraps. A station that stays silent for 3 consecutive turns is noted as dormant (it keeps costing only the Skip time until the mesh route TTL removes it). 911 alerts are the one exception to turn order: they may transmit in the next idle gap between turns after a short random delay.

Setting (Audio Setup tab)	Default	What it covers
Guard ms	1500	The radio's VOX hang after audio stops, plus margin
Skip ms	1200	VOX attack (~250 ms) + time to detect the carrier before declaring a station silent
Max turn ms	0 = automatic per modem (AFSK 6000, OFDM ~4400, Mercury ~6000)	The longest allowed batch; also the deadline timer

Radio profile — Baofeng UV-5R and Yaesu FT-65 (reference radios): neither radio's VOX hang is adjustable — the FT-65's VOX menu is ON/OFF only (Set Mode item 35, used with the SSM-512B earpiece mic) and the UV-5R exposes only VOX *sensitivity* 1–10 — and both hold the transmitter keyed roughly 1.0–1.5 s after audio stops. The 1500 ms Guard default covers that worst case. Their VOX attack also clips the first ~150–250 ms of audio, so set **VOX Lead = 250 ms** when driving these radios. Both radios' time-out timers (≥60 s) are far above the longest turn — no interaction.

Calibrating Guard: run a two-station session, key one radio with a short test transmission, and read the session log's receive timing (the gap between the end of decoded audio and the next burst, or the AFSK RX burst lines) to measure how long the carrier outlives the audio. Set Guard

= the worst measured hang among all radios in the net + 150 ms. Every station must use the same Guard, Skip and modem.

Capacity (20 stations, 5 of them with traffic in a given cycle, defaults): AFSK ≈ 32 s per full rotation, OFDM ≈ 35 s, Mercury ≈ 54 s. So that position reports never dominate the rotation, Ring mode automatically floors the PLI interval at $\text{roster} \times 8$ s ($\times 13$ s on Mercury), even if the user's "PLI min s" is set lower — that keeps PLIs at roughly a quarter of each station's turn capacity, leaving the rest for chat, ACKs and relays (raised from $\times 6$ s / $\times 10$ s in 0.23 after field testing).

ARQ interaction (plugin 0.23): the mesh layer's delivery retries (section 4.9) are aware of the Ring MAC. A 3-station field test on 0.22 showed the CSMA-era 5 s retry timer firing while the original Direct was still waiting for its turn: every chat message went to air up to 4 times, 20 of 23 Directs were declared failed even though they arrived, and TX queues peaked at ~21 frames (~4 minutes of backlog). Since 0.23: (1) a pending Direct's retry clock only starts once its latest copy actually *airs*; (2) under Ring the retry interval scales to at least $1.25\times$ the measured rotation time (an ACK cannot physically return sooner than the recipient's own turn); (3) chat delivered/read receipts travel as unacknowledged Directs; (4) a queued position report is replaced in place when a fresher one arrives; and (5) a persistent backlog is called out in the Activity Log (Ring: TX queue...). Raising **Max turn** (identically on every station) now also raises how many frames fit in one batch, so operators can trade longer turns for faster queue drain.

Design notes (why not the alternatives): REQUEST/GRANT schemes, RTS/CTS and explicit token packets all cost full modem bursts per exchange — on OFDM every burst is ~1.84 s and on Mercury ~5.6 s regardless of payload size, so control traffic is as expensive as data. A coordinator is also a single point of failure. Slotted TDMA wastes the slot remainder on short messages. The ring keeps the determinism of TDMA/token designs with none of their airtime overhead.

4. Mesh Network Layer

Implements “Adaptation of Uncoordinated Distance-Vector Routing for Unencrypted Amateur Radio Networks” (see References), with one approved deviation: packet payloads carry compact HBC binary rather than ITA2 text.

In plain terms: every station keeps a simple address book (“to reach DAVE, hand the message to BOB”). The address book is built just by listening, and messages for far-away stations get passed station-to-station like a relay race.

4.1 Common Packet Header (13 bytes)

Every mesh packet starts with the same 13 bytes:

Byte offset	Length	Field	Description
0	1	Packet Type	0x01 Announce, 0x02 Broadcast, 0x03 Direct, 0x04 ACK
1	5	Origin Callsign	Who originally created the packet (bit-packed ITA2, section 4.2)
6	5	Transmitter Callsign	Who is transmitting this particular hop; each relay overwrites it with its own callsign
11	2	Sequence ID	A 16-bit counter value (big-endian) used to recognize duplicates (section 4.8)

4.2 Callsign Encoding (ITA2, 5 bytes)

Callsigns are stored in an old teletype alphabet called ITA2, which uses only 5 bits per character instead of 8 — so 8 characters fit in exactly 5 bytes (40 bits). Two special “shift” codes switch between letters (LTRS = 11111) and figures/digits (FIGS = 11011); unused positions are filled with NULL (00000). A leading LTRS is written before the first letter; if a callsign would need 9 codes (for example KE8TQB), the leading LTRS is dropped — decoders start in letters mode anyway, so both forms read back identically.

Example	Code sequence	Packed bytes (hex)
K1ABC	LTRS K FIGS 1 LTRS A B C	FF B7 DF E2 6E
N3DEF	LTRS N FIGS 3 LTRS D E F	F9 B7 0F CA 16
W2XYZ	LTRS W FIGS 2 LTRS X Y Z	FE 77 9F DE B1

NOTE: The source paper's K1ABC example lists byte 03 as BF, but the paper's own bit-by-bit math works out to DF; this implementation follows the math. All three vectors above are locked in by the automated test suite (`codec-test/MeshRouterTest.java`).

4.3 Packet Type 0x01 — Announce (14 bytes)

A tiny “I exist and here’s how far away I am” packet that builds everyone’s address book.

Byte offset	Length	Field	Description
0–12	13	Common Header	Type = 0x01
13	1	Hop Count	How many relays the announce has passed through; the origin sets 0, each relay adds 1

4.4 Packet Type 0x02 — Broadcast (13 + n bytes)

A message to everyone in earshot. Not relayed, not acknowledged.

Byte offset	Length	Field	Description
0–12	13	Common Header	Type = 0x02
13+	variable	Payload	One HBC message (section 5)

4.5 Packet Type 0x03 — Direct Message (23 + n bytes)

A message for one specific station, passed hop-by-hop.

Byte offset	Length	Field	Description
0–12	13	Common Header	Type = 0x03
13	5	Destination Callsign	The final recipient
18	5	Next Hop Callsign	The neighbor being asked to carry this hop; each relay rewrites it
23+	variable	Payload	One HBC message

A station that sees its own callsign in Next Hop looks the Destination up in its address book, writes in the next station’s callsign, and re-transmits. The final destination delivers the payload

and answers with an ACK (section 4.6). If nobody knows a route, a relay drops the packet; the original sender instead falls back to sending it as a Broadcast.

4.6 Packet Type 0x04 — Acknowledgment (20 bytes)

The “got it!” receipt for a Direct message, sent back to the original sender.

Byte offset	Length	Field	Description
0–12	13	Common Header	Type = 0x04. Origin = the station acknowledging.
13	5	Destination Callsign	The Origin of the Direct being acknowledged
18	2	Acked Sequence ID	The Sequence ID of the Direct being confirmed

Stations along the way forward ACKs toward their destination when they know a route; the duplicate cache (section 4.8) keeps that from snowballing.

4.7 Route Discovery and Maintenance

- **Announce propagation** — when a station hears a new Announce, it writes the entry “Origin is reachable via whoever just transmitted this” and re-broadcasts the announce with hop count + 1. Shorter routes (fewer hops) replace longer ones; ties go to whichever route was heard first.
- **Passive learning** — hearing *any* valid frame is proof of a path, so the transmitting station is recorded as a direct neighbor, and (for non-announce frames) the originating station is recorded as reachable through that transmitter. Hearing a single position report is enough to be able to send to that station.
- **Announce-on-traffic** — a station’s own transmissions do the announcing for it; an explicit Announce packet only goes out after 10 quiet minutes, as a keepalive.
- **Expiration** — address-book entries are erased 30 minutes after the last frame that refreshed them (stations move and links change); a cleanup pass runs every 60 s.

4.8 Loop Prevention

Without protection, a rebroadcast message could bounce between stations forever. Each receiver therefore remembers [Origin + Sequence ID + Type] combinations for 5 minutes; a packet it has already seen is silently ignored. One deliberate exception: a duplicate Direct addressed *to this station* gets a fresh ACK (but is not delivered twice) — that covers the case where the first ACK was lost on the air.

4.9 Reliability and Retries

Directs work “fire and wait”: the originator keeps the message until the destination’s ACK arrives, re-sending it **with the same Sequence ID** up to 3 times, then declaring it failed and logging it. Broadcasts are never acknowledged or retried. Two refinements (plugin 0.23) let this ARQ coexist with the Ring MAC:

- **The retry clock starts when the frame airs, not when it is queued.** Under the Ring MAC a frame can wait many seconds for this station’s turn; retrying while the original is still queued only duplicates it in the same queue. A pending whose copies never reach the air at all is failed after 3 minutes (stuck-queue safety).
- **Retry pacing follows the channel-access mode.** CSMA: 5 s + 0–2 s random jitter (the original behavior). Ring: at least 1.25× the measured rotation time (minimum 8 s, capped at 60 s) — an ACK cannot return before the recipient’s own turn comes around, so earlier retries are pure overhead.

Chat delivered/read receipts (section 5.4) are sent as **unacknowledged Directs** (0.23): identical on the wire, but with no retries and no failure verdict — a lost checkmark does not justify retry traffic. The recipient still answers with a mesh ACK, which the origin simply ignores.

5. Application Layer (HBC Protocol v1.6)

The payload inside every Broadcast/Direct is one HBC message: a map event squeezed into as few bytes as possible by packing values bit-by-bit instead of sending bulky XML text. (For scale: a normal ATAK position report is roughly 1–2 *thousand* bytes of XML; the HBC version is about 15 bytes.) Full field-level definitions and worked byte examples live in the HBC-Protocol repository (see References); this section summarizes the interface.

5.1 HBC Header

Field	Size	Description
Callsign	variable (ITA2, CR-terminated)	Sending operator/device callsign, max 8 characters
Version	3 bits	000 = version 1
Mode	3 bits	mode-1 for Modes 1–6; 111 = Mode 0 Ack

NOTE: HBC uses its own ITA2 table (bit-reversed relative to the mesh header's table) and marks the end of text with a CR code instead of padding to a fixed size. The two layers' callsign encodings are cousins, not twins — they are not interchangeable.

5.2 Message Modes and Sizes

Mode	CoT type(s)	What it carries	Typical size
0 Ack	b-t-f-d / b-t-f-r	Chat delivered/read receipt: recipient callsign, kind (2 bits), 16-bit tag	~12 B
1 PLI / Spot	a-f-G, a-h-G, a-n-G, a-u-G	Position report: spot flag, affiliation (2 bits), name (7 chars), latitude/longitude (21+22 bits)	~15 B
2 Alert	b-a-o-tbl / b-a-o-can	911 emergency: active flag, alert + originator names, position	~18 B
3 GeoChat	b-t-f	Text message: destination kind (All / Room / DM), room or recipient, DM 16-bit tag, ITA2 text	DM “helo landing zone” = 23 B
4 Shape	u-d-c-c / u-d-r / u-d-f	Drawing: circle (center + radius) or polygon/polyline (≤15 points)	varies
5 CASEVAC	b-r-f-h-c	Medical evacuation request (9-line summary fields)	~20 B
6 Extended Marker	any placed marker	Full CoT type token, icon reference (2525C / spot-map / custom), color tint	varies

5.3 Deterministic UUIDs

ATAK identifies every map object by a UUID (unique ID string). HBC does not transmit UUIDs — both ends compute the same one from the callsign, so a station's position marker updates in place instead of piling up duplicates:

Object	UUID rule
Position (PLI) marker	HBC-{CALLSIGN}
911 alert	HBC-{CALLSIGN}-911
DM conversation	Recipient's UUID; the receiving plugin swaps in its real ATAK device UUID so the message lands in the right chat window
Spots / shapes / markers	Random UUID each time one is received

5.4 Direct-Message Tags and Receipts

Every chat DM carries a 16-bit tag — a CRC-16/CCITT-FALSE checksum (polynomial 0x1021, initial value 0xFFFF) of ATAK's internal `messageId`. When the recipient's ATAK automatically generates its delivered/read receipts, the plugin converts them into tiny Mode 0 Acks that echo the tag back over the air; the sender matches the tag to the original `messageId` and injects the receipt, so ATAK's familiar delivered/read checkmarks work end-to-end over radio. These application-level receipts are separate from (and in addition to) the mesh-layer ACKs of section 4.6. Since 0.23 they are carried as unacknowledged Directs (section 4.9) — best-effort by design.

5.5 Contact Publication

ATAK only lists a station in its chat/contact pickers when that station's position report includes a network "endpoint." Received PLIs are therefore reconstructed with `<contact callsign="..." endpoint="*:-1:stcp"/>` and a `<__group>` team entry, so hearing a single position report makes the station selectable for chat and file/point sends. Outgoing traffic to that contact is intercepted by the plugin and transmitted over the mesh.

6. User Interface Reference (Screenshots)

All configuration lives in one pane, opened from the HBC Radio entry in ATAK's Tools menu (Figure 1). This section shows each screen (6.1), explains every control (6.2) and toolbar icon (6.3), and teaches how to read the two logs (6.4, 6.5). Red arrows in the figures point at the controls each caption describes.

6.1 Screen-by-Screen Figures

NOTE (0.22): the figures below show the 0.20-era pane. Since 0.22 the pane is reorganized into three tabs — **Audio Setup** (modem, TX stream, AFSK level, channel access, timing), **Options** (callsign, Send-to, PLI rate, enables, message types) and **Decodes** (a full-panel page with its own ◀ Back button) — under a slim always-visible header carrying the colored status line and the Start/Stop button. The controls themselves are unchanged and are all described in section 6.2; refreshed screenshots will accompany the next figure update.

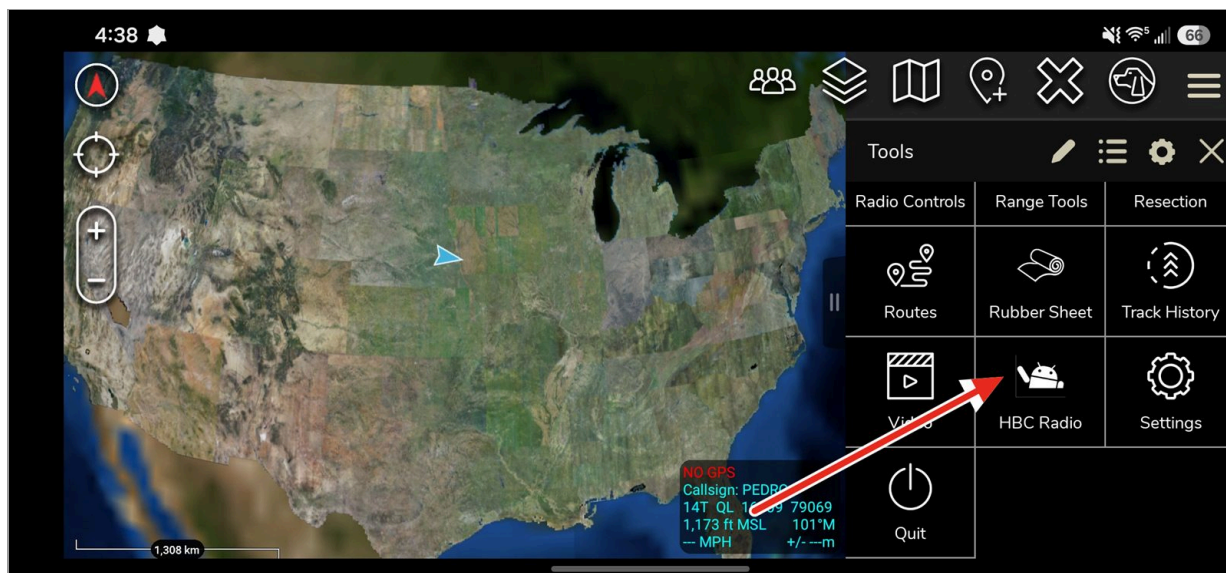


Figure 1 — Opening the plugin: ATAK menu (≡) → Tools → **HBC Radio** (arrow). The same grid is where ATAK lists every installed plugin.

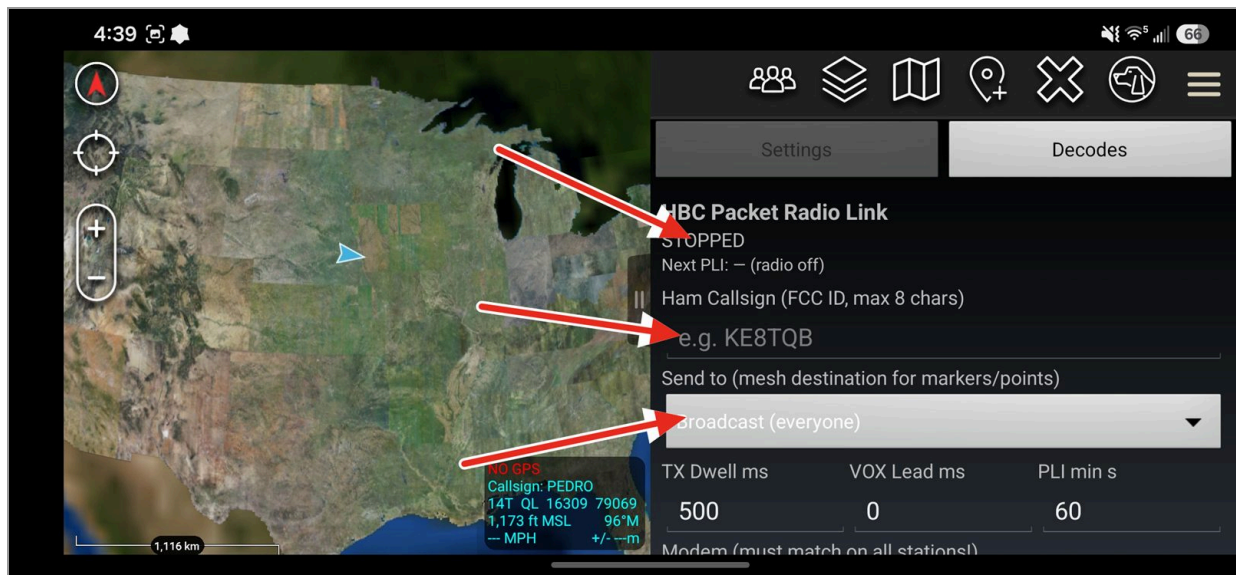


Figure 2 — The Settings tab, top half. Arrows, top to bottom: the link status line (STOPPED) with the “Next PLI” countdown beneath it; the ham callsign entry (required before starting); and the Send-to destination selector — “Broadcast (everyone)” by default, with every station learned by the mesh (section 4.7) added to the list automatically as a routed Direct destination for markers/points.

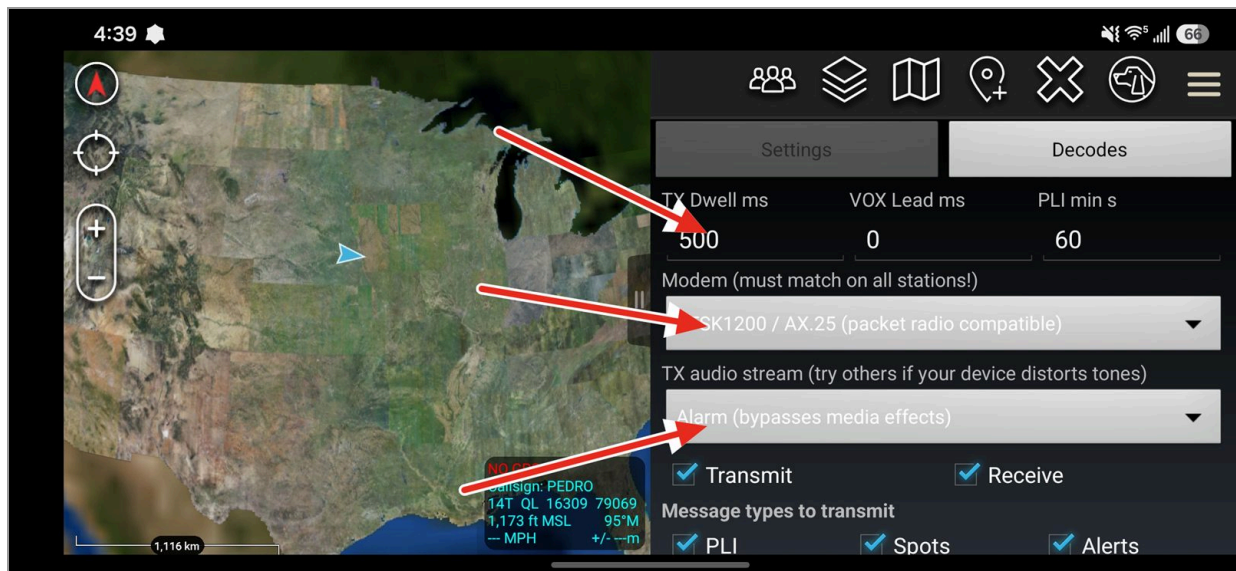


Figure 3 — Settings tab, middle. Arrows, top to bottom: the TX Dwell / VOX Lead / PLI rate timing fields (sections 3.1, 3.4, 4.7); the modem selector (must match on all stations — expanded in Figure 4); and the TX audio stream selector — “Alarm” is the default because it bypasses phone media effects (EQ, spatial audio) that can distort modem tones; Media, Ring, and Notification exist for devices that behave differently.

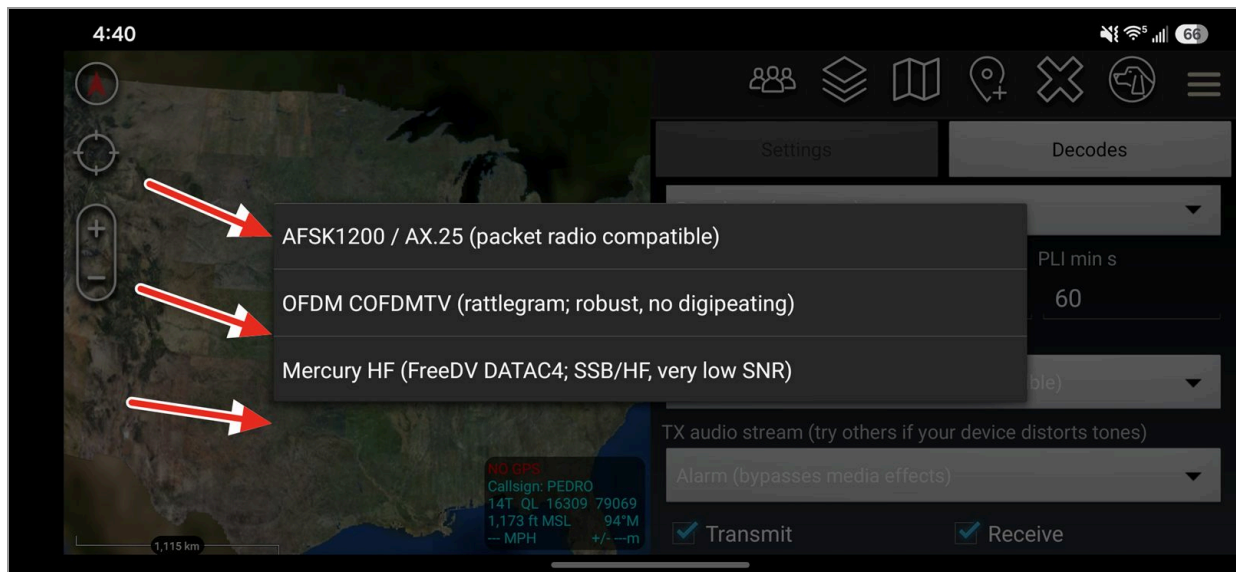


Figure 4 — The modem selector expanded (section 3), one arrow per waveform: AFSK1200/AX.25 (3.1), OFDM COFDM TV (3.2), and Mercury HF / FreeDV DATA4 (3.3). All stations on a channel must choose the same modem.

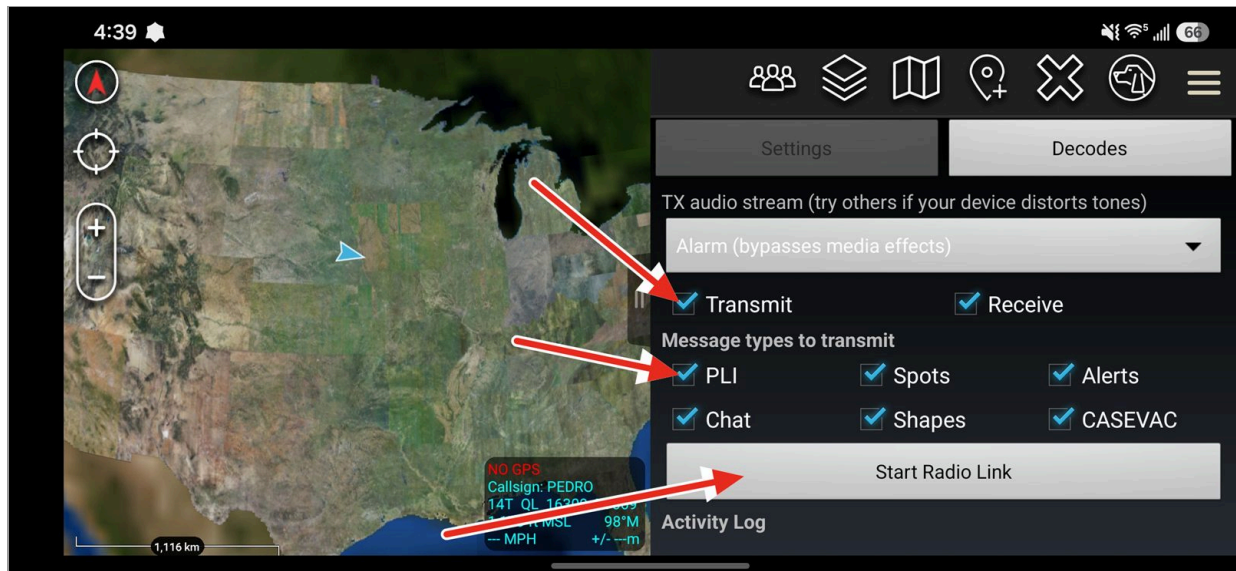


Figure 5 — Settings tab, bottom. Arrows, top to bottom: the Transmit/Receive master switches; the per-type transmit filters (PLI, Spots, Alerts, Chat, Shapes, CASEVAC — the HBC modes of section 5.2); and the Start Radio Link button. The Activity Log (section 6.4) scrolls beneath it.

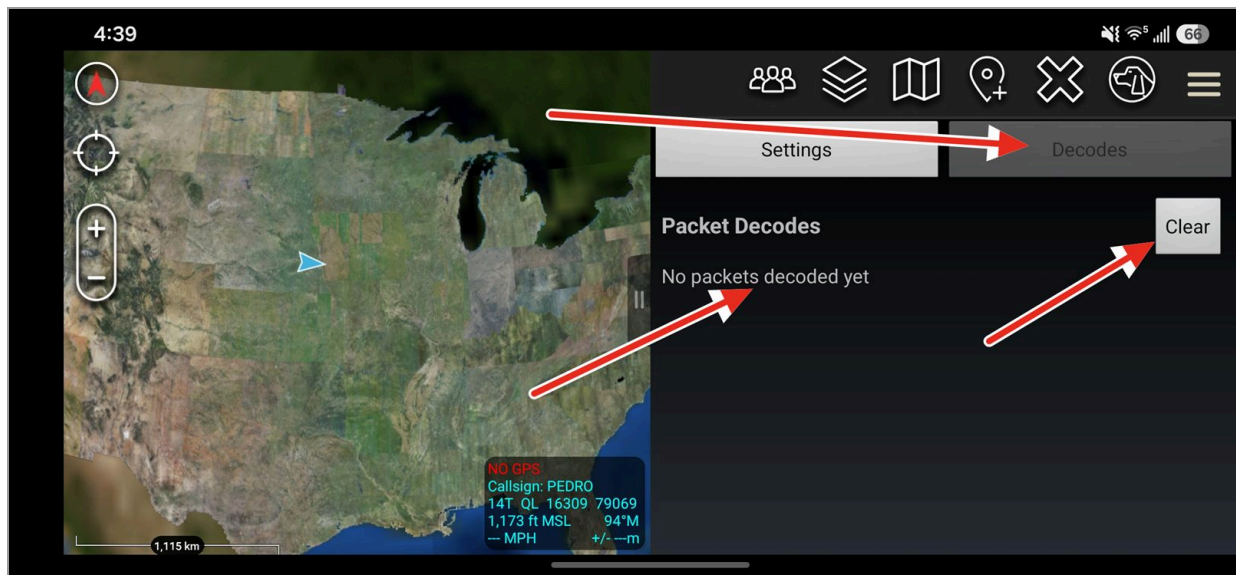


Figure 6 — The Decodes tab: a running log of every packet successfully received and decoded (empty here — nothing heard yet). Arrows: the active tab's button is the grayed one; Clear empties the log and resets its packet counter; decoded packets appear in the list area with sender, mode summary, and payload size (section 6.5).

6.2 Every Control and What It Does

Grouped by where each control lives in the 0.22 pane:

Control	Type	What it does
Audio Setup / Options / Decodes	Tab buttons	Switch screens; the active tab's button is grayed out. Decodes opens as a full-panel page (header and tab bar hidden) with its own ◀ Back button returning to the last settings tab.
Status line	Text (header, always visible)	HBC Radio – RUNNING as {callsign} in green while the modem is on; HBC Radio – STOPPED in red otherwise.
Next PLI / Build line	Live text (1 s updates)	While running: countdown until the next position report may transmit (in m:ss, ready (waiting for ATAK position update), or disabled). While stopped: Build {timestamp} · v{version} — use it to verify which binary is actually installed.
Ham Callsign	Text field	Your FCC callsign, max 8 characters (A–Z, 0–9, hyphen). Used as this station's mesh address and legal station ID. Required before starting.
Send to	Dropdown	Destination for markers/points you create: Broadcast (everyone), or any heard station for a routed, acknowledged Direct (section 4.5). Chat DMs ignore this and always route to their recipient automatically.

TX Dwell ms	Number field	AFSK only: length of the HDLC flag preamble sent before data so the receiving modem can sync. Default 500. Increase if receivers miss the start of packets.
VOX Lead ms	Number field	Warm-up audio played before data so a VOX-keyed radio has time to key up. Default 0 (use 0 with a hardwired PTT interface). Increase if the first part of transmissions is cut off.
PLI min s	Number field	Minimum seconds between your own position reports (rate limit). Default 60. Drives the “Next PLI” countdown.
Modem	Dropdown	Which of the three waveforms to use (section 3). Listed OFDM (Rattlegram), Mercury, AX.25/APRS; OFDM is the default . Must match on every station.
TX audio stream	Dropdown	Which Android audio channel carries the transmit tones (Figure 3). Default Alarm; try the others only if your device distorts the tones.
Channel access	Dropdown	CSMA (legacy contention, section 3.4) or Ring (deterministic turns, section 3.5). Must match on every station.
① UV-5R / FT-65 defaults	Help link	Opens a dialog with the recommended Ring/VOX values for the reference radios, why their VOX hang cannot be shortened, the Guard calibration procedure, and hardwired-PTT alternatives (section 3.5).
Guard ms / Skip ms / Max turn ms	Number fields	Ring MAC timing (section 3.5): Guard covers the radios’ VOX hang (default 1500), Skip covers VOX attack before skipping a silent station (default 1200), Max turn caps one batch (0 = automatic per modem). Same values on every station. Since 0.23 a larger Max turn also fits more frames into each batch (faster queue drain at the cost of longer turns).
Transmit / Receive	Checkboxes	Master switches. Unchecking Transmit makes the station receive-only; unchecking Receive ignores all incoming audio.
PLI, Spots, Alerts, Chat, Shapes, CASEVAC	Checkboxes	Per-type transmit filters, matching the HBC modes of section 5.2. An unchecked type is never transmitted (receiving is unaffected).
Start / Stop Radio Link	Button (header, always visible)	Validates and saves the settings, then starts (or stops) the selected modem and the mesh layer. The label flips to match the current state.
Activity Log	Scrolling text	Timestamped diagnostic feed of everything the plugin does (section 6.4). Newest entries on top.
Clear (Decodes tab)	Button	Empties the Decodes log and resets its packet counter.

6.3 Toolbar Icons

Icon	Tap	Notes
HBC Radio (radio-tower glyph)	Opens the plugin pane	The plugin's single entry point in ATAK's Tools menu (Figure 1). The Tools grid tints icons white, so the entry uses a transparent-background tower silhouette; the colored badge appears as the app icon in TAK Package Mgmt and the plugin manager.

Changed in 0.22: the separate gray/green radio-status toolbar icon (tap-to-toggle) was removed — the radio link is controlled entirely from the pane's Start/Stop button, and the link state is shown by the colored header status line (red STOPPED / green RUNNING).

6.4 Reading the Activity Log

Every line starts with a HH:mm:ss timestamp; the newest line is at the top. The tables below list the messages verbatim (`{braces}` mark values filled in at runtime) and what each one means.

Radio link and modem:

Message	Meaning
Radio link started (<code>{modem}</code> , mesh routing)	Start succeeded; the named modem and the mesh layer are running.
Radio link stopped	The modem was stopped (button, icon tap, or plugin shutdown).
Start failed: <code>{error}</code>	The modem could not start (usually a microphone/audio problem). The link is left stopped.
Modem started ... / RX audio source: <code>{source}</code>	Which microphone source the receiver opened (VOICE_RECOGNITION, UNPROCESSED, or MIC; “via USB” when a Digirig-style interface is attached).
RX disabled: microphone unavailable	ATAK lacks the RECORD_AUDIO permission or another app holds the mic; receiving is off.
TX <code>{details}</code> (<code>{n}</code> ms burst)	A transmission finished playing; <code>n</code> is its on-air duration.
CSMA: channel busy – deferring TX	Someone else is on the air; the plugin is politely waiting (section 3.4).
CSMA: channel busy > 8 s – transmitting anyway	The channel never cleared, so the frame was sent to avoid starving traffic.
OFDM sync: <code>{call}</code> ... / Mercury sync acquired	The demodulator locked onto an incoming signal (decode result follows).

TX batch: {n} frames ({t} ms burst)	Several queued frames went out as one continuous keying (one VOX cycle for the whole batch).
--	--

Channel access (Ring MAC, section 3.5):

Message	Meaning
Ring: started ({n} stations, guard {g} ms, skip {s} ms)	Ring mode is on with these parameters and this roster size.
Ring: joined rotation as {call} ({n} stations)	The listening (settle) period is over; this station now takes its turns.
Ring: TX turn ({i}/{n}, {x} frames)	It is this station's turn and {x} queued frames were released as one batch.
Ring: roster now {n} stations ({list})	The rotation list changed at a cycle wrap (station joined or expired).
Ring: {call} dormant (3 silent turns)	That station has not transmitted for 3 cycles; it keeps costing only the Skip time.
Ring: emergency frame queued – preempting at next idle window / Ring: emergency TX ({x} frames, preempt)	A 911 alert jumped the turn order in an idle gap (the single allowed exception).
Ring: TX queue {n} frames (~{t} s to drain)	More than 6 frames are waiting for this station's turns (estimate from the measured rotation time; logged at most every 30 s). If it persists, slow down chat/marker sends — see section 3.5's ARQ-interaction note.

Transmit decisions:

Message	Meaning
TX candidate: {type} uid {uid}	ATAK asked to send this event; the plugin is deciding whether to transmit it.
Queued TX mode {n} ({x} B) {type}	The event was encoded as HBC mode n (x bytes) and handed to the mesh for transmission.
Queued TX mode 0 ack ({type}) -> {call} tag 0x{tag}	A chat delivered/read receipt is being sent back to a DM sender (section 5.4).
TX skip: type {type} disabled in settings	Dropped: that message type's checkbox is off.
TX skip: PLI rate limit	Dropped: a position report arrived before "PLI min s" elapsed (see the Next PLI countdown).

TX skip: duplicate within 10 s	Dropped: ATAK re-sent the same event; the copy was suppressed to save airtime.
TX skip: {reason}	Dropped: the event has no HBC representation (e.g. text with unsupported characters, coordinates out of range).

Mesh routing:

Message	Meaning
Mesh: route {call} via {call2} ({n} hops)	Address-book update: {call} is now reachable by handing frames to {call2}. A trailing [heard] means it was learned passively just by hearing traffic (section 4.7). The Send-to list refreshes.
Mesh: announce sent (seq {n})	The 10-minute quiet-time keepalive announce went out.
Mesh: direct to {call} via {call2} (seq {n})	A routed Direct left this station; an ACK is now awaited.
Mesh: direct to {call} via {call2} (seq {n}, unacked)	A fire-and-forget Direct (chat delivered/read receipt): no retries, no failure verdict (section 4.9).
Mesh: {call} acknowledged seq {n}	Delivery confirmed — the destination's ACK arrived.
Mesh: retry {i}/3 to {call} via {call2}	No ACK yet; the Direct was re-sent (same sequence ID). Since 0.23 this fires only after the previous copy actually aired — under Ring, at least ~1.25 rotations later.
Mesh: direct to {call} FAILED (no ack after 3 retries)	Delivery gave up. The station is likely out of range or off. Also issued when queued copies could not reach the air for 3 minutes (stuck queue).
Mesh: no route to {call} – sending as broadcast	No address-book entry for the destination, so the message went out unrouted; the station will still get it if in direct range.
Mesh: forwarding {call} - > {call2} via {call3}	This station is acting as a relay for someone else's Direct.
Mesh: no route to {call} – direct dropped	Asked to relay a Direct but no route to the destination is known.

Receive decisions:

Message	Meaning
RX {call} {summary}	A packet was decoded and injected into ATAK (the same event also appears in the Decodes tab, section 6.5).

RX chat: DM for '{call}' – not this station, ignored	A direct chat addressed to someone else was overheard and correctly not shown.
RX ack for '{call}' – not this station, ignored	Same, for a chat receipt.
RX ack tag 0x{tag} – no matching sent DM, ignored	A receipt arrived for a message this station has no record of sending (e.g. app restarted).
RX decode produced invalid CoT ({summary})	The packet decoded but ATAK rejected the rebuilt event — report this as a bug.

6.5 Reading the Decodes Log

The Decodes tab (Figure 6) lists only *successfully decoded* packets — a clean record of what actually arrived. The header shows a running total ({n} packet(s) decoded; Clear resets it). Each entry has three lines:

```
[{MM-dd HH:mm:ss}] {origin callsign}
    {message summary}
    {n} B payload
```

The summary line identifies the HBC mode and its key contents:

Summary format	How to read it
Mode 1 PLI {call} '{name}' @ {lat},{lon}	Position report from {call}; the marker {name} was placed/updated at those coordinates. Spot instead of PLI means a placed marker, not a self-position.
Mode 2 Alert(ACTIVE) {name} @ {lat},{lon}	A 911 emergency from {name} at that location. (CANCEL) means the emergency was called off.
Mode 3 Chat {call}: {text}	A chat message and its text (broadcast, room, or DM — DMs for other stations are never injected).
Mode 0 Ack(DELIVERED) {call} -> {call2} tag 0x{tag}	{call} confirms a DM from {call2} was delivered; (READ) means it was opened. Produces the chat checkmarks.
Mode 4 Shape({k}) {call} '{name}'	A drawing (0 = circle, otherwise polygon/polyline) named {name}.
Mode 5 CASEVAC {call} '{name}'	A medical-evacuation request titled {name}.
Mode 6 Marker {type} {call} '{name}' @ {lat},{lon}	A full marker with its exact CoT type and icon (section 5.2, Mode 6).

Rule of thumb: the Activity Log answers “what is the plugin doing and why?” (including everything it decided *not* to send); the Decodes log answers “what have I actually received?”

7. Timing Parameters and Constants

Parameter	Value	Layer
Route entry TTL	30 min (refreshed by any heard frame)	Mesh
Duplicate cache TTL	5 min	Mesh
Cleanup interval	60 s	Mesh
Direct retry timer	Armed when the frame airs (0.23). CSMA: 5 s + 0–2 s jitter; Ring: max(8 s, 1.25 × measured rotation), cap 60 s. Max 3 retries; stuck-queue fail 180 s	Mesh
Quiet-time announce keepalive	10 min without own traffic	Mesh
CSMA busy threshold / hold	4× adaptive noise floor / 400 ms	PHY
CSMA backoff / give-up	150–550 ms clear / 8 s (15 s Mercury)	PHY
Ring Guard (VOX hang cover)	User setting, default 1500 ms (UV-5R / FT-65 profile)	MAC
Ring Skip (silent-station timeout)	User setting, default 1200 ms	MAC
Ring Max turn	Auto per modem: AFSK 6000 / OFDM 4400 / Mercury 6000 ms (user-overrideable)	MAC
Ring settle (join listening)	1 observed cycle, max 30 s	MAC
Ring batch caps	Derived from Max turn (0.23); at the automatic defaults: AFSK 4 / OFDM 2 / Mercury 1 frames per keying (also applies under CSMA)	MAC/PHY
PLI rate limit	User setting, default 60 s (Figure 2); Ring mode floors it at roster × 8 s (× 13 s Mercury, raised in 0.23)	Application
RX duplicate window (frame hash)	30 s	Plugin

8. Compatibility and Constraints

- All stations on a channel must run the same plugin wire version (0.12+ mesh framing) and select the same modem (Figure 4).

- The channel-access mode (section 3.4/3.5) is **behavioral only** — nothing on the air changes — so 0.21-and-earlier stations interoperate with 0.22 stations. For orderly rotation every station should still select the same mode and the same Guard/Skip values; a CSMA station mixed into a Ring net simply contends as before and the Ring stations defer to its carrier (the net degrades toward polite CSMA, never below it).
- The 0.23 ARQ changes (TX-gated, rotation-scaled retries; unacknowledged receipts; PLI floor $\times$ 8 s) are likewise behavioral only — wire-identical with 0.22. In a mixed net, 0.22-and-earlier stations simply keep their old 5 s retry behavior (and its queue growth) while 0.23 stations do not.
- Mercury DATAC4's small frame limits routed Directs to ~29 bytes of HBC payload (Broadcasts ~39 B). Every standard message fits; unusually large Mode 6 markers may need Broadcast or a different modem.
- Callsigns: at most 8 ITA2 characters (A–Z, 0–9, hyphen).
- No encryption is used or permitted; never transmit sensitive data.

9. Glossary

Term	Meaning
ATAK	Android Team Awareness Kit — the mapping app this plugin extends
CoT	Cursor on Target — the XML message format ATAK uses for map events
EUD	End User Device — the phone/tablet running ATAK
PLI	Position Location Information — a “here I am” report
Modem	Modulator/demodulator — converts bytes to sound and back
Frame / packet	One complete chunk of bytes sent over the air
Payload	The useful data inside a packet, after the addressing header
Hop	One radio link in a relay chain (A→B is one hop; A→B→C is two)
Route / next hop	Address-book entry: which neighbor to hand a message to so it reaches a destination
ACK	Acknowledgment — a short “received OK” reply
ARQ	Automatic Repeat reQuest — send, await the ACK, re-send on timeout (section 4.9)
CSMA	Carrier Sense Multiple Access — listen before transmitting to avoid collisions
MAC	Medium Access Control — the rules deciding when a station may transmit (CSMA or Ring here)
Ring MAC / turn	The 0.22 channel-access mode: stations rotate through the sorted roster; a turn is one station's transmit window

Guard time	Silence a Ring station waits after the channel clears, covering the previous radio's VOX hang
VOX	Voice-operated transmit — the radio keys up automatically when it hears audio
CRC	Cyclic Redundancy Check — a checksum that detects corrupted data
ITA2	A 5-bit teletype alphabet from the telegraph era; smaller than normal 8-bit text
TTL	Time To Live — how long a record is kept before being erased
UID	Unique identifier string for a map object in ATAK

10. References

1. HBC Protocol reference implementation and specification — github.com/ke8tqb/HBC-Protocol
2. HBC ATAK Plugin source — github.com/ke8tqb/HBC-ATAK-Plugin
3. *Adaptation of Uncoordinated Distance-Vector Routing for Unencrypted Amateur Radio Networks* (routing protocol paper)
4. Mercury HF modem — mercury.hermes.radio; FreeDV/codec2 — github.com/drowe67/codec2
5. rattlegram / COFDMTV — github.com/aicodix/rattlegram

Appendix A — GitHub Repository File Structure

The plugin source lives at github.com/ke8tqb/HBC-ATAK-Plugin. The map below shows where everything described in this document is implemented.

HBC-ATAK-Plugin/	
├─ app/	
│ └─ build.gradle	Plugin version, ATAK version, NDK pin, signing config
│ └─ proguard-gradle.txt	Release-build shrinker rules
│ └─ src/main/	
│ └─ AndroidManifest.xml	Plugin discovery entry required by ATAK
│ └─ java/	
│ └─ com/atakmap/android/hbc/	
│ └─ HbcEncoder.java	Application layer: CoT XML → HBC bytes (Section 5)
│ └─ HbcDecoder.java	Application layer: HBC bytes → CoT XML
│ └─ Ita2.java	HBC's 5-bit text alphabet (Section 5.1)
│ └─ BitReader.java	Bit-level read helper for the codec
│ └─ BitWriter.java	Bit-level write helper for the codec
│ └─ MeshRouter.java	Mesh network layer: packets, routing, retries, ACKs (Section 4)
│ └─ CsmaSense.java	Listen-before-transmit logic (Section 3.4)
│ └─ AudioModem.java	AFSK1200 modem (Section 3.1)
│ └─ OfdmModem.java	OFDM COFDMTV modem (Section 3.2)
│ └─ OfdmNative.java	Java ↔ C++ bridge for the OFDM library
│ └─ MercuryModem.java	Mercury HF modem (Section 3.3)
│ └─ MercuryNative.java	Java ↔ C bridge for the FreeDV library
│ └─ plugin/	
│ └─ HbcPlugin.java	The plugin itself: settings pane, toolbar icons, TX/RX plumbing, ATAK integration
│ └─ sivantoledo/ax25/...	Bundled javAX25 library (AFSK/AX.25, GPL)
├─ cpp/	
│ └─ CMakeLists.txt	Native build: libhbcofdm + libhbcmercury
│ └─ hbc-ofdm-jni.cpp	C++ glue for the OFDM modem
│ └─ *.hh	Vendored aicodix OFDM/DSP headers
│ └─ hbc-mercury-jni.cpp	C glue for the FreeDV DATAC modem
│ └─ hbc-mercury-compatible.c	Math shims for older Android versions
│ └─ mercury/	Vendored FreeDV/codec2 sources (from the Mercury project; LGPL – see THIRD-PARTY.md)
├─ res/	
│ └─ layout/main_layout.xml	The settings/decodes pane (Figures 1–6)
│ └─ values/strings.xml	All UI text and dropdown options
│ └─ drawable/	Icons, incl. ic_radio_on/off (status icon)
├─ codec-test/	
│ └─ HbcCodecTest.java	Byte-exact tests for the HBC codec (runs on a PC)
│ └─ MeshRouterTest.java	Byte-exact + behavior tests for the mesh layer
├─ docs/	
│ └─ HBC_ICD.html	This document (editable source)
│ └─ HBC_ICD.pdf	This document (rendered)
│ └─ images/	Figures 1–6 (app screenshots)
└─ gradle/, gradlew, gradlew.bat	Gradle build system wrapper

└─ settings.gradle, build.gradle	Project-level build configuration
└─ .takdev/	TAK SDK build-plugin support files
└─ CHANGELOG.md	Version-by-version history of changes
└─ THIRD-PARTY.md	Licenses of all bundled third-party code
└─ README.md	Project introduction and build instructions

Where to look first: the wire formats of Section 4 are all in `MeshRouter.java`; the message encodings of Section 5 are in `HbcEncoder.java/HbcDecoder.java`; anything you can tap in the app is in `HbcPlugin.java` plus `res/layout/main_layout.xml`. The codec-test/ programs run on an ordinary computer (no phone needed) and prove the byte formats match this document.

HBC Radio ICD Version 1.2 | Aug 2026 | KE8TQB | github.com/ke8tqb/HBC-ATAK-Plugin